Ubiquinone Biosynthesis from Chorismate-Derived 4-Hydroxybenzoate in *Pseudomonas putida* KT2440

Commissioned Module/Pathway/Taxon Review — KEGG ppu00130 (cofactors_vitamins_redox) Target taxon: *Pseudomonas putida* KT2440 (PSEPK; NCBI taxon 160488; proteome UP000000556)

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1. Executive Summary

The de novo ubiquinone (coenzyme Q, in this strain **UQ-9**) biosynthetic pathway from the chorismate-derived aromatic head group **4-hydroxybenzoate (4-HB)** is **fully satisfiable** in *P. putida* KT2440. Every canonical step of the bacterial Ubi pathway maps to a conserved, syntenically plausible gene in the KT2440 proteome: chorismate lyase (**UbiC**, PP_5317), 4-HB polyprenyltransferase (**UbiA**, PP_5318), the prFMN-dependent decarboxylation module (**UbiX**, PP_0548, supplying prenylated-FMN to **UbiD**, PP_5213), the three aerobic ring hydroxylations (**VisC/UbiI**, PP_5197; **UbiH**, PP_5199; and a **Coq7-type** enzyme, PP_0427, substituting for the flavin-dependent UbiF that is absent from this genome), and the two ring methylations (O-methyltransferase **UbiG**, PP_1765; C-methyltransferase **UbiE**, PP_5011). The polyprenyl (nonaprenyl) side chain is supplied by IspB (PP_0687), which lies outside the KEGG map.

Two curation-critical corrections to the supplied metadata are warranted. **First, the candidate list is incomplete**: it omits three accessory/regulatory factors that are present in the proteome and are integral to *Pseudomonas*/proteobacterial UQ biosynthesis — the atypical kinase **UbiB** (PP_5013), and the lipid-binding assembly factors **UbiJ** (PP_5012) and **UbiK** (PP_5235). Notably, *ubiE–ubiJ–ubiB* form a contiguous cluster (PP_5011–PP_5012–PP_5013), strong syntenic support for their functional linkage. **Second, the candidate list is inflated** by five map-boundary artifacts that are not part of de novo UQ biosynthesis from 4-HB: *hpd* (PP_3433, a homogentisate/tyrosine-catabolism dioxygenase belonging to the tocopherol–plastoquinone branch), the acyl-CoA/4-hydroxybenzoyl-CoA thioesterase PP_1218, and three

NAD(P)H:quinone oxidoreductases (PP_1644, PP_2789, PP_3720) that *use* quinones rather than *synthesize* them. These arise because KEGG ppu00130 is the lumped map "Ubiquinone **and other terpenoid-quinone** biosynthesis."

The single most important caveat for curators is that **all 14 candidate annotations are homology-inferred (UniProt PE=3)**, with no protein- or transcript-level experimental evidence in KT2440 itself. The pathway is confidently reconstructed by orthology, KEGG KO assignment, and strong transfer from the well-characterized *Escherichia coli* and *Pseudomonas aeruginosa* systems, but not by direct KT2440 biochemistry or genetics. A published genome-wide RB-TnSeq fitness resource for KT2440 offers the fastest route to convert these homology calls into strain-specific functional evidence.

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2. Target-Organism Pathway Definition

Included process. This review covers **de novo aerobic biosynthesis of ubiquinone** starting from **chorismate** (via 4-hydroxybenzoate) and a **polyprenyl diphosphate** side chain, through to mature UQ-9. The chemically defined scope is: (i) 4-HB formation from chorismate; (ii) prenylation of 4-HB; (iii) prFMN-dependent decarboxylation (with UbiX supplying the cofactor to UbiD); (iv) three contiguous aromatic-ring hydroxylations; and (v) the O- and C-methylations that decorate the ring. In *P. putida* KT2440 the isoprenoid tail is **nonaprenyl (C45)**, giving UQ-9, consistent with other pseudomonads.

Neighboring pathways to keep separate. KEGG map **ppu00130 is deliberately broad** — "Ubiquinone *and other terpenoid-quinone* biosynthesis" — and co-locates enzymes for **menaquinone**, **tocopherol**, **phylloquinone**, and **plastoquinone**. For a KT2440-specific UQ module these must be excluded: - **Menaquinone (MK)** biosynthesis: not present in KT2440 (proteome searches for *menA* and *menD* return zero hits), consistent with a strict aerobe that respires exclusively via UQ. - **Tocopherol/plastoquinone branch**: the *hpd* gene (4-hydroxyphenylpyruvate dioxygenase, PP_3433) belongs to **tyrosine catabolism / homogentisate** metabolism, not de novo UQ from 4-HB. - **Chorismate biosynthesis (shikimate pathway)** upstream and **quinone *utilization*** (respiratory NAD(P)H:quinone oxidoreductases) downstream should be treated as adjacent pathways, not part of this module.

Alternate names / database definitions. The pathway is variously called ubiquinone biosynthesis, coenzyme Q biosynthesis, and (in this strain) UQ-9 biosynthesis. Gene-name synonymy that curators must reconcile includes **VisC** = **UbiI** (the C5/2-methoxy-6-

polyprenylphenol hydroxylase; "VisC" is the historical name) and the substitution of a **Coq7-type hydroxylase (EC 1.14.99.60)** for the reaction otherwise assigned to the flavin monooxygenase **UbiF (EC 1.14.13.-)** in *E. coli*.

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3. Expected Step Model

The species-neutral outline maps onto KT2440 genes as follows. The pathway topology (which step feeds which) is preserved; only the enzyme identity at the C6-hydroxylation step differs from the *E. coli* paradigm.

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Chorismate
   │  UbiC (PP_5317)  chorismate lyase (EC 4.1.3.40)
   ▼
4-Hydroxybenzoate  ───+ nonaprenyl-PP (from IspB / PP_0687, outside map)
   │  UbiA (PP_5318)  4-HB polyprenyltransferase (EC 2.5.1.39)
   ▼
3-nonaprenyl-4-hydroxybenzoate
   │  UbiD (PP_5213)  prFMN-dependent decarboxylase (EC 4.1.1.98)
   │  UbiX (PP_0548)  supplies prenylated-FMN cofactor (EC 2.5.1.129)
   ▼
2-nonaprenylphenol
   │  three CONTIGUOUS aromatic ring hydroxylations (aerobic):
   │    VisC/UbiI (PP_5197)  C5 hydroxylase (EC 1.14.13.240)
   │    UbiH      (PP_5199)  C6 hydroxylase
   │    Coq7-type (PP_0427)  C6/DMQ hydroxylase (EC 1.14.99.60)  ← replaces UbiF
   │  and two methylations:
   │    UbiG (PP_1765)  O-methyltransferase (EC 2.1.1.222 / 2.1.1.64) — acts twice
   │    UbiE (PP_5011)  C-methyltransferase (EC 2.1.1.163/2.1.1.201)
   ▼
Ubiquinone-9 (UQ-9)

Accessory / regulatory (MISSING from candidate metadata, present in proteome):
   UbiB (PP_5013)  atypical kinase / assembly & CoQ distribution
   UbiJ (PP_5012)  SCP2 lipid-binding assembly factor (Ubi metabolon)
   UbiK (PP_5235)  assembly factor (partners UbiJ)
   [ubiE–ubiJ–ubiB are a contiguous cluster: PP_5011–PP_5012–PP_5013]

NOT expected in KT2440 (strict aerobe):
   UbiT / UbiU / UbiV  O2-independent hydroxylation system — ABSENT (0 hits)
   UbiF                flavin C6-hydroxylase — ABSENT genome-wide
   MenA / MenD         menaquinone — ABSENT
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Step satisfiability summary table:

Expected step	Enzyme family	KT2440 gene	Call
4-HB from chorismate	UbiC chorismate lyase	PP_5317 (*ubiC*)	**covered**
Polyprenylation of 4-HB	UbiA prenyltransferase	PP_5318 (*ubiA*)	**covered**
prFMN cofactor supply	UbiX flavin prenyltransferase	PP_0548 (*ubiX*)	**covered**
Decarboxylation	UbiD prFMN decarboxylase	PP_5213 (*ubiD*)	**covered**
Ring hydroxylation (C5)	UbiI/VisC	PP_5197 (*visC*)	**covered**
Ring hydroxylation (C6)	UbiH	PP_5199 (*ubiH*)	**covered**
Ring hydroxylation (C6/DMQ)	Coq7-type (replaces UbiF)	PP_0427 (*coq7*)	**covered (lineage substitution)**
O-methylation (×2)	UbiG	PP_1765 (*ubiG*)	**covered**
C-methylation	UbiE	PP_5011 (*ubiE*)	**covered**
Side-chain synthesis	IspB nonaprenyl-PP synthase	PP_0687	**covered (outside map)**
Assembly/regulation	UbiB kinase	PP_5013 (*ubiB*)	**covered — ADD to metadata**
Assembly	UbiJ	PP_5012 (*ubiJ*)	**covered — ADD to metadata**
Assembly	UbiK	PP_5235 (*ubiK*)	**covered — ADD to metadata**
Anaerobic hydroxylation	UbiT/U/V	—	**not_expected_in_target_taxon**

4. Candidate Genes and Evidence

All 14 supplied candidates share one over-arching caveat established in **Finding F002**: every UniProt entry carries **proteinExistence = "3: Inferred from homology"**. None is PE=1 (protein-level) or PE=2 (transcript-level). This means the annotations rest on sequence orthology and pathway-database placement, not on KT2440-specific experiments.

4.1 High-confidence biosynthetic core (promote all to full review)

UbiC — PP_5317 (Q88C66), chorismate pyruvate-lyase, EC 4.1.3.40; KO K03181. Catalyzes the entry reaction, chorismate → 4-HB + pyruvate. KEGG KO and EC are unambiguous. Curation note: UniProt labels it "Probable," reflecting homology-only evidence.

UbiA — PP_5318 (Q88C65), 4-HB octaprenyltransferase, EC 2.5.1.39; KO K03179. Prenylates 4-HB with the nonaprenyl tail. Directly adjacent to *ubiC* (PP_5317/PP_5318), a classic proteobacterial arrangement supporting the assignment. High confidence.

UbiX — PP_0548 (Q88QE6), flavin prenyltransferase, EC 2.5.1.129; KO K03186. Synthesizes the prenylated-FMN (prFMN) cofactor required by UbiD. Curation caveat: its **primary bucket in the metadata is kegg:ppu00627** (aminobenzoate/other), not ppu00130 — i.e., it is annotated to a different map yet is functionally essential to this pathway. The UbiX–UbiD partnership is one of the best-characterized modules in the field (multiple mechanistic studies; see §8), so transfer to KT2440 is strong.

UbiD — PP_5213 (Q88CG8), 3-octaprenyl-4-hydroxybenzoate decarboxylase, EC 4.1.1.98; KO K03182. The prFMN-dependent (de)carboxylase. High confidence. Caveat: the UbiD family is large and functionally diverse (it also includes aromatic-acid decarboxylases such as phenazine-1-carboxylic acid decarboxylase, PMID: 38176649); annotation transfer within the family requires the UbiX partner and pathway context, both of which are satisfied here.

VisC/UbiI — PP_5197 (Q88CI4), FAD/NAD(P)-binding oxidoreductase, EC 1.14.13.240; KO K18800. The C5 (2-methoxy-6-polyprenylphenol) hydroxylase. "VisC" is the historical name for UbiI. Adjacent to *ubiH* (PP_5197/PP_5199). The UniProt free-text ("Oxidoreductase involved in anaerobic synthesis of ubiquinone") is **misleading** — this is an aerobic hydroxylase; curators should not read "anaerobic" here as evidence for the O2-independent UbiT/U/V system (which is absent).

UbiH — PP_5199 (Q88CI2), 2-octaprenyl-6-methoxyphenol hydroxylase; KO K03185. The C6 hydroxylase. High confidence; syntenic with *visC*.

Coq7-type — PP_0427 (Q88QR1), 3-demethoxyubiquinol 3-hydroxylase / DMQ hydroxylase, EC 1.14.99.60; KO K06134. This is a **lineage-specific substitution**: KT2440 lacks the flavin monooxygenase UbiF (EC 1.14.13.-; KO K18801 absent genome-wide, Finding F007) and instead uses a **Coq7/COQ7-family di-iron hydroxylase** (the enzyme normally associated with eukaryotic/alphaproteobacterial CoQ synthesis) to accomplish the equivalent C6 hydroxylation. This is a known evolutionary theme: proteobacteria deploy varied enzymes with different regioselectivities to accomplish the three contiguous hydroxylations (PMID: 27822549). High curation value — flag as a **non-canonical but real** assignment.

UbiG — PP_1765 (Q88M10), UQ O-methyltransferase, EC 2.1.1.222 / 2.1.1.64; KO K00568. Performs **both** O-methylations in the pathway (bifunctional, acts at two steps). High confidence.

UbiE — PP_5011 (Q88D17), C-methyltransferase, EC 2.1.1.163/2.1.1.201; KO K03183. The ring C-methyltransferase. High confidence. Caveat: UniProt/HAMAP appends a **generic "menaquinone biosynthesis; menaquinol... step 2/2"** pathway string to UbiE. Because KT2440 has **no menaquinone pathway** (*menA/menD* absent), this menaquinone annotation is an **over-propagated artifact** — UbiE here functions only in UQ biosynthesis. First member of the *ubiE–ubiJ–ubiB* cluster.

4.2 Accessory factors present but MISSING from the candidate list (Finding F001)

UbiB — PP_5013 (A0A140FWS4), "Probable protein kinase UbiB," EC 2.7.-.-; KO K03688; HAMAP MF_00414. Atypical ABC1/ADCK/UbiB-family kinase required for efficient CoQ biosynthesis and for CoQ distribution across membranes (functionally characterized in *E. coli*, yeast Coq8, human ADCK3/4; PMID: 34362905, PMID: 32479562, PMID: 18319074). Present in the proteome, contiguous with *ubiJ* and *ubiE*. **Should be added to the module.**

UbiJ — PP_5012 (Q88D16), UbiJ accessory factor; HAMAP MF_02215. SCP2 lipid-binding protein that scaffolds the cytosolic Ubi "metabolon" and binds pathway intermediates (PMID: 36142227). Directly between *ubiE* and *ubiB*. **Add to the module.** No KEGG KO is assigned (Finding F007), so it is invisible to KO-based reconstruction — a systematic reason it was dropped.

UbiK — PP_5235 (Q88CE6), UbiK accessory factor; HAMAP MF_02216. Partners UbiJ in the assembly complex. Present in proteome; no KEGG KO. **Add to the module.**

4.3 Map-boundary artifacts — likely NOT part of de novo UQ (Findings F004, F005)

Gene	ID / accession	Annotation	Why it is an artifact
hpd	PP_3433 (Q88HC7)	4-hydroxyphenylpyruvate dioxygenase, EC 1.13.11.27, KO K00457	Tyrosine catabolism / homogentisate → tocopherol–plastoquinone branch, **not** UQ from 4-HB
PP_1218	Q88NI9	acyl-CoA / 4-hydroxybenzoyl-CoA thioesterase, EC 3.1.2.23, KO K01075	Peripheral hydrolase; not a UQ biosynthetic step
PP_1644	Q88MC9	WrbA-type NAD(P)H:quinone oxidoreductase, EC 1.6.5.2, KO K03809	**Uses** quinones (respiration/ stress), does not make UQ
PP_2789	Q88J60	NAD(P)H:quinone oxidoreductase (Qor/MdaB), EC 1.6.5.2, KO K00355	Quinone-utilizing, not biosynthetic
PP_3720	Q88GK1	NAD(P)H:quinone oxidoreductase (Qor/MdaB), KO K00355	Quinone-utilizing, not biosynthetic

These five inflate the candidate list purely because KEGG ppu00130 is a multi-quinone overview map. They should be excluded from a KT2440-specific de novo UQ module.

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5. Gaps, Ambiguities, and Likely Over-Annotations

5.1 The candidate list = raw KEGG ppu00130 membership. A KEGG REST query (`link/ppu/path:ppu00130`) returns exactly the 14 PP_ IDs supplied (Finding F005). This confirms the metadata was generated by direct map membership, which explains both its omissions (accessory factors *ubiJ/ubiK* have no KO; *ubiB* and *ispB* are not map members) and its inflation (multi-quinone map artifacts).

5.2 UbiF → Coq7 substitution (Findings F003, F007). Genome-wide KO checks confirm **no UbiF ortholog** (K18801 absent) and **no UbiI-independent flavin C6-hydroxylase**. The three aerobic hydroxylations are covered by VisC/UbiI, UbiH, and the Coq7-type enzyme. This is consistent with the established evolutionary flexibility of proteobacterial UQ hydroxylases — *"The biosynthesis of UQ requires three hydroxylation reactions on contiguous positions of an aromatic ring"* (PMID: 27822549). Curators should record the Coq7 assignment as a **confirmed lineage substitution**, not a gap.

5.3 Anaerobic UbiT/U/V not expected (Finding F006). No *ubiT/ubiU/ubiV* genes (0 hits). In denitrifying *P. aeruginosa* PAO1 this O2-independent system is essential for anaerobic UQ-9 synthesis (PMID: 32409583); a *P. aeruginosa* genome-scale model (iSD1509) added it because the pathway *"is required for anaerobic growth"* (PMID: 36765199). KT2440 is a **strict aerobe lacking a complete denitrification chain**, so this branch should be marked **not_expected_in_target_taxon** rather than a gap. This is a clear case where a generic multi-organism module boundary would be **wrong** for KT2440.

5.4 Over-propagated annotations to flag. - **UbiE menaquinone pathway string** — spurious in KT2440 (no *men* genes). - **VisC "anaerobic" free-text** — misleading; it is an aerobic hydroxylase. - **Broad EC/GO on the quinone oxidoreductases** (EC 1.6.5.2, KO K00355/K03809) — these map to respiration/detox, not biosynthesis.

5.5 Homology-only evidence (Finding F002). The deepest gap is evidential, not genomic: **all** calls are PE=3. No KT2440 protein-level, transcript-level, or mutant-phenotype data anchor any step in the strain itself.

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6. Module and GO-Curation Recommendations

Step disposition for the module:

Module element	Recommended status
UbiC, UbiA, UbiX, UbiD, VisC/UbiI, UbiH, UbiG, UbiE	**covered** (homology-strong)
Coq7-type C6 hydroxylation (PP_0427)	**covered** — annotate as lineage substitution for UbiF
Nonaprenyl-PP side chain (IspB PP_0687)	**covered** — note it lies outside the KEGG map
UbiB (PP_5013), UbiJ (PP_5012), UbiK (PP_5235)	**covered** — **module_needs_revision to ADD these**
hpd, PP_1218, PP_1644, PP_2789, PP_3720	**remove** — map-boundary artifacts, not de novo UQ
UbiT/U/V anaerobic hydroxylation	**not_expected_in_target_taxon**
UbiF flavin C6-hydroxylase	**not_expected** (absent; role filled by Coq7)

Module boundary guidance. The generic module is broadly correct in topology but must be **species-tailored** for KT2440: (i) exclude menaquinone/tocopherol/plastoquinone members inherited from the lumped KEGG map; (ii) explicitly encode the **Coq7-for-UbiF** substitution as an accepted alternative enzyme; (iii) add the accessory-factor sub-step (UbiB/UbiJ/UbiK); and (iv) mark the anaerobic branch as not expected.

GO-term / annotation requests. Because **UbiJ and UbiK lack KEGG KOs**, KO-driven module reconstruction will systematically miss them — a GO/annotation request to ensure these HAMAP-annotated accessory factors (MF_02215/MF_02216) are captured in the module ontology would prevent recurrent under-calling across proteobacteria. No new GO terms appear strictly necessary for the core enzymatic steps, which are well covered by existing EC/GO mappings.

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7. Genes to Promote to Full `fetch-gene` Review

Prioritized for individual review because they are either lineage-specific substitutions, missing from metadata, or carry misleading annotations:

1. **PP_0427 (Coq7-type)** — confirm it, not a UbiF ortholog, performs the C6 hydroxylation; highest novelty/curation value.

2. **PP_5013 (UbiB)**, **PP_5012 (UbiJ)**, **PP_5235 (UbiK)** — add to module; verify identity and synteny (*ubiE–ubiJ–ubiB* cluster).
3. **PP_5197 (VisC/UbiI)** — correct the misleading "anaerobic" free-text; confirm C5-hydroxylase role.
4. **PP_5011 (UbiE)** — strip the spurious menaquinone pathway annotation.
5. **PP_0548 (UbiX)** — reconcile its primary bucket (ppu00627) with its essential UQ role.
6. **PP_5213 (UbiD)** — verify family assignment given the functionally diverse UbiD superfamily.
7. **PP_0687 (IspB)** — confirm as the nonaprenyl-PP source feeding UbiA (outside the KEGG map).

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8. Evidence Base and Key References

Direct KT2440 evidence: essentially none at the pathway level — this is the central limitation. The most actionable strain-specific resource is the **RB-TnSeq fitness library** disrupting ~4,778 non-essential genes of KT2440 (PMID: 33964456), part of the Deutschbauer/Arkin Fitness Browser (Finding F008). Genes *absent* from such non-essential libraries are inferred essential; core aerobic-respiration UQ genes are expected to be essential or strongly fitness-defective aerobically. A related RB-TnSeq study of *P. putida* carbon metabolism (PMID: 32826213) demonstrates the approach is mature for this strain.

Mechanistic transfer from related systems (strong):

Topic	PMID	How it supports the review
Three contiguous UQ hydroxylations; variable enzymes	27822549	Underpins the three-hydroxylation requirement and acceptance of Coq7-type enzymes at these positions
O2-independent UbiT/U/V pathway	32409583	Defines the anaerobic branch that KT2440 lacks
P. aeruginosa genome-scale model iSD1509	36765199	Anaerobic UQ-9 pathway "required for anaerobic growth" — not-expected in aerobic KT2440
UbiX–UbiD / prFMN biochemistry	32951834, 31142738, 28754323, 38176649	Cofactor chemistry and UbiD family diversity; justify pathway-context requirement for UbiD annotation
UbiJ–UbiK assembly factors	36142227	Supports adding accessory factors to the module
ABC1/ADCK/UbiB kinase family	34362905, 32479562, 18319074, 24267661	Supports adding UbiB (PP_5013) as an accessory/regulatory factor

Verified supporting quotes. - *"The biosynthesis of UQ requires three hydroxylation reactions on contiguous positions of an aromatic ring."* (PMID: 27822549) - *"we newly added a recently discovered pathway for ubiquinone-9 biosynthesis which is required for anaerobic growth."* (PMID: 36765199)

Database evidence (direct for KT2440 genome content): UniProt proteome UP000000556 gene-name and proteinExistence queries (Findings F001, F002); KEGG REST map membership and KO/EC assignments (Findings F005, F007). These are authoritative for gene presence/absence and annotation provenance but are themselves homology-based for function.

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9. Limitations and Knowledge Gaps

1. **No direct target-strain functional data.** Every functional call is PE=3 homology inference. Confidence in the *reconstruction* is high; confidence in *strain-specific biochemistry* is homology-grade.

2. **Coq7 substitution is inferred, not demonstrated in KT2440.** The absence of UbiF plus presence of a Coq7 ortholog with the matching EC strongly implies substitution, but the actual C6-hydroxylation activity in KT2440 has not been shown experimentally.
3. **Accessory-factor essentiality unknown in KT2440.** UbiB/UbiJ/UbiK roles are transferred from *E. coli*/yeast/*P. aeruginosa*; their quantitative contribution in KT2440 is untested.
4. **Side-chain length assumed UQ-9.** Consistent with pseudomonads and IspB annotation, but not directly measured here.
5. **KEGG map lumping** forces manual boundary decisions; the five artifact genes and the menaquinone/tocopherol overlap require curator judgment rather than automated exclusion.

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10. Proposed Follow-up Experiments / Actions

Curation actions (immediate): - Revise the module to **add** PP_5013 (UbiB), PP_5012 (UbiJ), PP_5235 (UbiK) and **remove** the five artifacts (PP_3433, PP_1218, PP_1644, PP_2789, PP_3720). - Record the **Coq7-for-UbiF** substitution and mark **UbiT/U/V** as not_expected_in_target_taxon. - Correct misleading free-text on VisC ("anaerobic") and the spurious menaquinone pathway string on UbiE. - File an annotation/GO request so KO-less accessory factors (UbiJ/UbiK) are captured in future reconstructions.

Experimental / data-mining actions (to close the PE=3 gap): - **Mine the KT2440 RB-TnSeq Fitness Browser** (PMID: 33964456) for per-gene fitness/essentiality of *ubiA/C/D/X/G/E*, *visC*, *ubiH*, *coq7*, *ubiB/J/K* under aerobic growth — the single fastest path to strain-specific evidence. - **Targeted deletion + UQ-9 quantification** (LC-MS) for PP_0427 (Coq7) to confirm the UbiF-substitution hypothesis; a Δ*coq7* strain should accumulate demethoxy-UQ. - **Complementation of an *E. coli ubiF* mutant** with KT2440 *coq7* to demonstrate the C6-hydroxylation activity heterologously. - **Verify UQ-9 chain length** by mass spectrometry and confirm IspB (PP_0687) as the nonaprenyl-PP source.

Report generated from a 5-iteration autonomous review integrating UniProt (UP000000556), KEGG (ppu00130) REST queries, and 23 literature records. All gene presence/absence claims derive from direct database queries of the KT2440 proteome; all functional roles are homology-inferred unless otherwise stated.